# EXPERIMENT 10
# TURBULENT PIPE-FLOW MODEL COMPARISON

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## 1. Aim

To compare radial eddy-viscosity closures for fully developed turbulent pipe flow by examining mean velocity profiles, Darcy friction factors, pressure-drop implications and the axial velocity field.

## 2. Governing equation and numerical model

The supplied solver assumes steady, incompressible, axisymmetric and fully developed pipe flow. The mean axial velocity varies only with radial position, U = U(r), and the radial and circumferential mean velocities are zero. The axial momentum equation with an effective viscosity is:

$$\mathbf{0 = -dp/dx + (1/r)\, d/dr\, [\, r\, (\nu + \nu_t)\, dU/dr\, ]}$$

Here, $\nu$ is the molecular kinematic viscosity and $\nu_t$ is the turbulent (eddy) kinematic viscosity. The effective viscosity is $\nu eff = \nu + \nu_t$. The five model options are laminar, mixing-length, Van-Driest-damped mixing-length, equilibrium k–ε and equilibrium k–ω/SST surrogate. The k–ε and k–ω/SST options are algebraic equilibrium closures in this teaching solver, not full transport-equation RANS models.

## 3. Boundary conditions and assumptions

- At the pipe centreline, r = 0: symmetry condition, dU/dr = 0.
- At the pipe wall, r = R: no-slip condition, U = 0.
- The bulk mean velocity is normalized to the target value $\bar{U}b = 1$.
- The pipe radius is R = 0.5 and the diameter is D = 1.0 in the solver coordinate system.
- The flow is treated as fully developed, with a constant axial pressure gradient.

## 4. Run parameters and data availability

The supplied figures identify the displayed case as Re = 10,000. The friction-factor plot contains one Reynolds-number location ($10^4$), and the velocity-profile plot is explicitly titled for pipe2d_profiles_Re10000. Therefore, the discussion and numerical estimates in this report are restricted to the displayed Re = 10,000 case.

The screenshots do not display the run-mode selection, radial grid size, VTK grid size, iteration limit, tolerance, or the solver CSV values. These settings cannot be verified from the figures alone. For reference, the notebook presets are:

| Setting | quick preset | lab preset |
|---|---|---|
| Radial points | 21 | 120 |
| VTK grid (x × y) | 30 × 21 | 180 × 60 |
| Reynolds range | 10,000–12,000 | 10,000–100,000 |
| Number of Re cases | 1 | 3 |
| Maximum iterations | 8 | 1,500 |
| Tolerance | $1 \times 10^{-4}$ | $1 \times 10^{-7}$ |

Important: because the supplied friction plot shows only Re = 10,000, it is consistent with the quick preset, but the screenshots alone do not prove which preset was selected. Do not report the lab grid or lab convergence settings as measured settings unless confirmed from the notebook cells or output logs.

## 5. Velocity-profile comparison at Re = 10,000

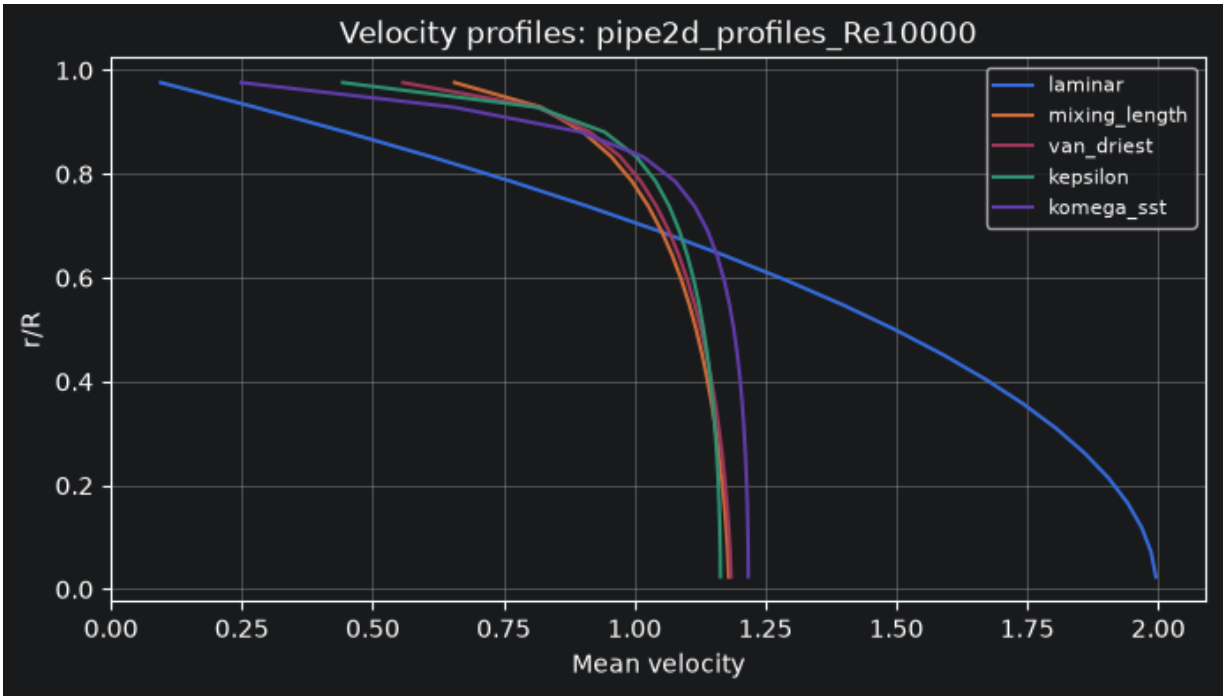



Figure 1. Mean axial velocity profiles for the five model options at Re = 10,000.

The laminar profile is parabolic: it has the highest centreline velocity (approximately 2 in the plotted normalized units) and decreases strongly toward the wall. The turbulent-model curves are flatter across the central region and lie close together around the core velocity. This is the fuller-profile behaviour associated with turbulent momentum mixing. In the plotted case, the k–ω/SST surrogate curve is the furthest to the right through much of the core, while the mixing-length and Van-Driest curves are close to one another. Near the wall, the model curves separate modestly; the graph should be used for qualitative comparison rather than claiming precise point values.

## 6. Darcy friction-factor comparison

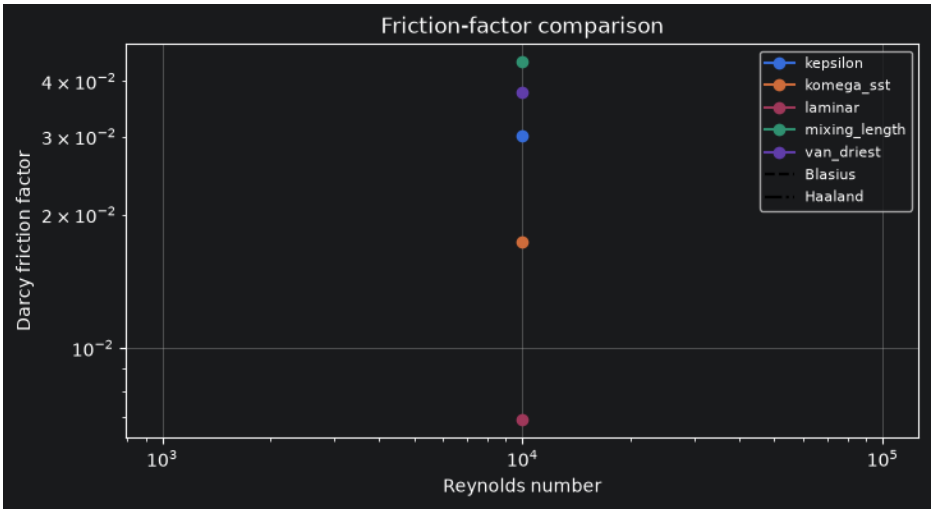



Figure 2. Darcy friction-factor comparison at Re = 10,000.

Approximate values read from the plotted markers are summarized below. They are visual estimates from the image, not substitutes for the exact values in pipe2d_rans_summary.csv.

| Model / reference | Approx. Darcy friction factor | Observation |
|---|---|---|
| Laminar | 0.0069 | Lowest plotted value |
| k–ω/SST surrogate | 0.017 | Lowest of the turbulent closures |
| k–ε | 0.030 | Close to smooth-pipe references |
| Van Driest | 0.038 | Above smooth-pipe references |
| Mixing length | 0.044 | Highest plotted value |
| Blasius reference | 0.03164 | Correlation value at Re = 10,000 |
| Haaland smooth reference | 0.03089 | Correlation value at Re = 10,000 |

The Blasius value is calculated from $f_D = 0.3164\,Re^{-0.25}$. The smooth-pipe Haaland value is calculated from $f_D = [-1.8\log_{10}(6.9/Re)]^{-2}$. At Re = 10,000, the k–ε marker is close to both reference correlations. The mixing-length and Van-Driest markers are higher, while the k–ω/SST surrogate is lower. The laminar marker is much lower, as expected when no turbulent eddy viscosity is included. Since the plot shows a single Reynolds-number point, it does not establish how the models trend with Reynolds number.

## 7. Inline velocity-field visualization

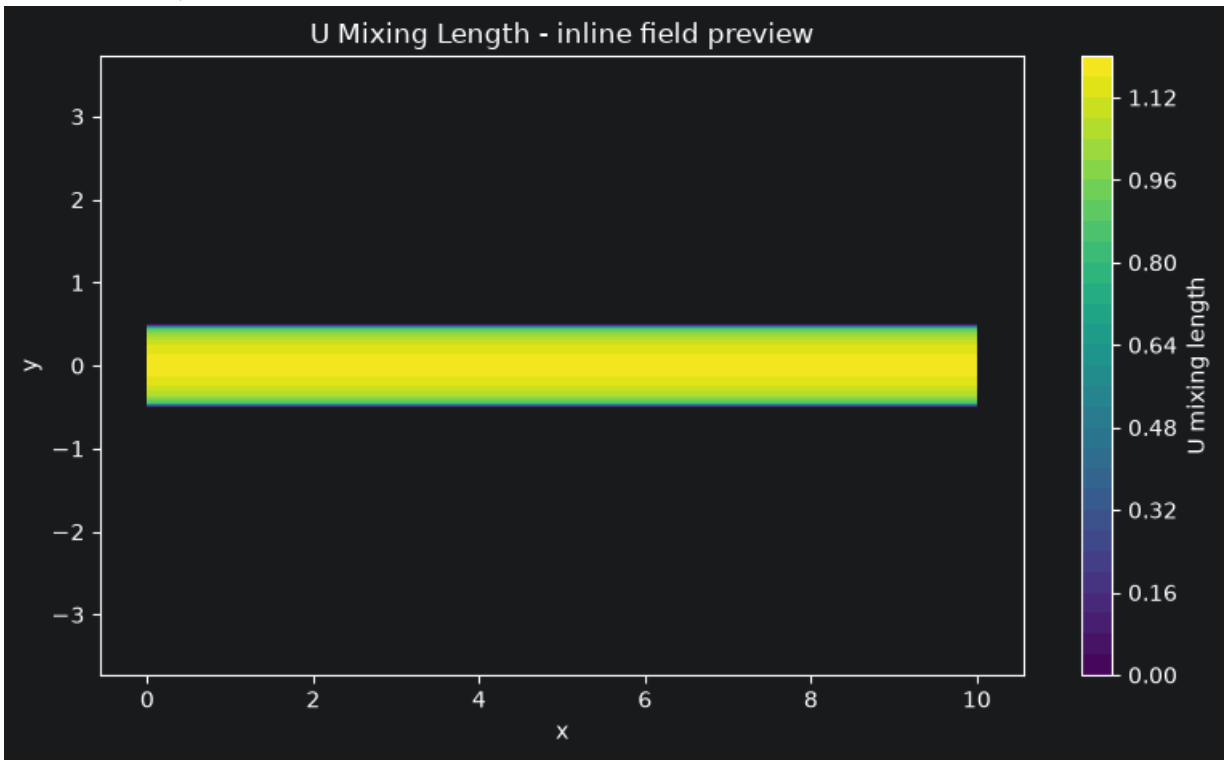



Figure 3. Inline contour of axial velocity U for the mixing-length closure at Re = 10,000. The horizontal axis is the streamwise coordinate x and the vertical axis is the transverse coordinate y.

The contour shows a high-velocity region around the pipe centre and lower velocity near the upper and lower walls. The field remains essentially unchanged along x, which is consistent with the fully developed assumption. This is an extruded radial profile for visualization; it is not evidence of a separately solved two-dimensional developing-flow field.

## 8. Pressure drop and turbulent viscosity

The solver calculates Darcy friction factor from $f_D = 8(u_\tau/\bar{U}_b)^2$. For a pipe length L, the Darcy–Weisbach relation is $\Delta p = f_D\,(L/D)(\rho \bar{U}_b^2/2)$. The notebook summary is designed to report pressure drop over 10D. However, the supplied screenshots do not show numerical pressure-drop values or turbulent-viscosity distributions. Therefore, exact pressure-drop and maximum $\nu_t/\nu$ values are not stated in this report. They should be copied from the generated summary CSV if those values are required by the evaluator.

## 9. Residual or error history

The solver defines the relative velocity-update residual as:

$$\text{Residual} = \|U^n - U^{n-1}\|_2 / \max(\|U^{n-1}\|_2, 10^{-30})$$

The code stores the final residual and iteration count in the summary CSV, but it does not write a residual value for every iteration. The supplied three plots do not contain residual data. Consequently, a residual-versus-iteration history and exact final residuals cannot be reported from these screenshots. They must be obtained by saving residual values inside the solver iteration loop or by using the generated summary CSV for the final residual only.

## 10. Observed numerical behaviour and physical interpretation

### Observed from the supplied plots

- The laminar velocity profile is parabolic and has a centreline value close to 2.
- The four turbulent-model profiles are flatter through the core than the laminar profile.
- The mixing-length model has the highest plotted Darcy friction factor, approximately 0.044.
- The k–ω/SST surrogate has the lowest friction factor among the turbulent closures, approximately 0.017.
- The k–ε friction-factor point is close to the Blasius and Haaland smooth-pipe reference values.
- The inline velocity contour is nearly uniform along the axial direction and varies mainly across the pipe.

### Physical interpretation

Turbulent eddies transport axial momentum from the faster central region toward the slower near-wall region. This radial momentum exchange makes the mean turbulent velocity profile fuller than the laminar parabolic profile. The different closures estimate turbulent viscosity differently, so they produce different wall gradients, friction factors and pressure-drop predictions. A larger friction factor indicates a larger wall-friction loss for the same pipe length, diameter and bulk velocity. These results compare the supplied teaching closures; they do not prove that one full turbulence model is universally more accurate.

## 11. Limitations of the teaching model

- The calculation is steady, axisymmetric and fully developed; entrance and developing-flow effects are not resolved.
- The k–ε and k–ω/SST options are equilibrium/algebraic surrogates, not full two-equation RANS transport simulations.
- The 2D-looking field is formed by extruding the radial profile along the pipe axis.
- The eddy viscosity is clipped at 300ν, which may affect the resulting profiles and friction factors.
- Only the final residual and iteration count are written; the complete residual history is not saved.
- The supplied screenshots show only Re = 10,000, so trends across the full Reynolds-number range cannot be concluded from them.
- Exact pressure-drop and turbulent-viscosity numerical values are not visible in the supplied figures.

## 12. Conclusion

At Re = 10,000, the supplied results show that turbulent closures produce fuller mean-velocity profiles than the laminar model. The mixing-length closure gives the highest plotted friction factor, followed by Van Driest, while k–ε is close to the Blasius and Haaland smooth-pipe references. The k–ω/SST surrogate gives the lowest friction factor among the turbulent closures in this plotted case. The inline contour confirms the expected centre-high, wall-low velocity distribution and its streamwise invariance under the fully developed assumption. These conclusions are limited to the displayed case and the simplified equilibrium radial-closure teaching model.